

# Warnings, clean care, and emergency access

Three papers · two development-focused general-interest reads and one EMS access note. The following article pages are edited reading excerpts, not complete replacements for the originals.

## LITERATURE PULSE

This issue's common thread is **nominal coverage is not operational access**. Disaster warnings save lives only when poorer and less-connected households receive and can act on them. Clinics cannot deliver safe care without water, sanitation, waste management, and cleaning - basic inputs whose absence creates a large hidden health and economic burden across African countries. Rural ambulance systems show the same last-mile problem in a richer setting: formal coverage can coexist with response times nearly twice as long. Development policy often succeeds or fails in the final mile.

01 RECENT / GENERAL INTEREST · STARTS ON PAGE 2 · World Bank Policy Research Working Paper 11199 · 2025 · CC BY 3.0 IGO

## Early Warnings Coverage, Determinants of Reception, and Benefits: What Surveys Can Tell Us

Combining the World Risk Poll's roughly 147,000 respondents in 142 countries with household surveys and Indonesian village data, the authors ask who receives disaster warnings and what happens afterward. Coverage rises with national income, while education, dwelling location, and digital access shape reception within countries. Receiving a warning is associated with more preventive action and higher perceived resilience; villages with warning systems have casualty rates estimated at roughly one-fifth those without them. These are strong policy-relevant associations, but the largely observational designs do not fully eliminate selection into warning coverage.

02 GENERAL INTEREST / GLOBAL HEALTH · STARTS ON PAGE 17 · World Bank Policy Research Working Paper 10708 · 2024 · CC BY 3.0 IGO

## Costs of Health Care Associated Infections from Inadequate Water and Sanitation in Health Care Facilities in Eastern and Southern Africa

A cost-of-illness model for Angola, the Democratic Republic of Congo, Ethiopia, Kenya, Madagascar, Mozambique, Sudan, Tanzania, and Uganda estimates 3.1 million health care-associated infections and more than 320,000 excess deaths in 2022. The modeled economic burden exceeds \$6 billion, or 1.14 percent of combined GDP, while basic water, sanitation, hygiene, waste management, and cleaning investments yield an estimated benefit-cost ratio of 5.8. The scale is striking, but the estimates inherit sparse surveillance data, cross-country extrapolation, and assumptions about mortality, length of stay, and the value of statistical life.

03 EMS / ACCESS · STARTS ON PAGE 32 · Research in Health Services & Regions · 2024 · CC BY 4.0

## Socioeconomic factors influencing rural-urban ambulance response-time disparities in Connecticut

Using town-level response times from Connecticut's 2016 EMS report, the study compares rural and non-rural places and tests whether median household income explains the gap. Rural mean response time is almost 13 minutes - roughly twice the non-rural mean - yet income is not significantly correlated with response time within either group. This points toward geography, stationing, staffing, and volunteer-service capacity rather than town income alone. Treat the result as descriptive: the data are old, town-level, and cross-sectional, so they cannot isolate causal mechanisms or within-town inequality.

*Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.*

# Early Warnings Coverage, Determinants of Reception, and Benefits

## What Surveys Can Tell Us

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## Abstract

Early warning systems are a cornerstone of disaster risk reduction, enabling timely alerts and protective actions that save lives and reduce damages. Yet, despite their documented cost-effectiveness and potential to bolster resilience, early warning system coverage remains uneven across countries and within communities. Furthermore, there remains a gap in the literature around people-centered approaches to understand what drives early warning system coverage, the reception of warnings, and their realized benefits at the household level. This paper addresses that gap by leveraging four surveys conducted at different scales (international, national, and subnational) and focusing on different geographic areas (urban versus rural) to examine early warning systems from a household perspective. Using the World

Risk Poll, the paper first examines the determinants of early warning system coverage at the national level, highlighting the role of gross domestic product, disaster predictability, and the likelihood that a country is affected by a disaster. At the household level, the findings show that warning reception is influenced by socioeconomic status, dwelling location, and access to digital devices. Finally, the paper shows that receiving an early warning is strongly associated with higher perceived resilience, increased adoption of preventive measures, and significantly lower casualty rates in disaster-affected areas. These findings underscore the urgent need to improve both the accessibility and effectiveness of early warning systems, particularly for vulnerable populations in low-income and high-risk regions.

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This paper is a product of the Urban, Disaster Risk Management, Resilience and Land Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://www.worldbank.org/prwp>. The authors may be contacted at [pavner@worldbank.org](mailto:pavner@worldbank.org) and [bjafino@worldbank.org](mailto:bjafino@worldbank.org).

*The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.*

## 1. Introduction

Disasters pose a significant challenge to global socioeconomic development. Over the past decade, geophysical, climate, and weather-related disasters have collectively resulted in economic damages averaging approximately US\$170 billion annually (UNDRR, 2022). Disaster impacts are unevenly distributed, with developing countries and smaller economies experiencing more pronounced short-term macroeconomic declines after disasters of comparable relative magnitude with respect to developed nations or larger economies (Noy, 2009). Looking at specific hazards, 23% of the global population faces substantial risk from 1-in-100-year flood events, with 170 million of these individuals living in extreme poverty (Rentschler et al., 2022). Floods have been the most frequent natural disasters causing up to 1,000 fatalities per event over the past 50 years, with flash floods accounting for the highest mortality rates per event (Jonkman et al., 2024). Tropical cyclones also pose a significant threat, with an estimated 560 million people exposed annually between 2002 and 2019, a trend that has increased across all cyclone intensities during this period (Jing et al., 2024). In aggregate, a World Bank assessment indicates that climate change, through various disaster and productivity channels, could result in an additional 100 million people in extreme poverty absent good, risk-informed, development policies (Hallegatte et al., 2016). As such, reducing disaster risks is a crucial component for achieving sustained global socioeconomic development.

Early warning systems (EWS) are one critical non-structural measure for disaster risk reduction. A well-designed EWS integrates a series of interconnected systems that monitor, analyze, and disseminate data on impending threats, facilitating timely alerts for at-risk populations. These systems rely on a strong technological foundation, including effective observation networks, advanced data processing and modeling techniques, and efficient communication and dissemination platforms. Additionally, the success of an EWS depends on sound decision-making frameworks that ensure the timely translation of forecast information into actionable warnings. This includes setting clear thresholds for alert issuance, coordinating responses across relevant stakeholders, and ensuring that the warnings are communicated in a way that is accessible and comprehensible to the public. To maximize effectiveness, communities receiving these warnings must have pre-established preparedness plans in place.

In the context of the traditional framework that expresses risk as the product of hazard, exposure, and vulnerability, EWS plays a key role in mitigating the exposure of vulnerable populations and their assets to potential hazards. By providing early alerts, these systems help people move themselves and some of their belongings out of harm's way. Additionally, EWS contribute to adaptive capacity by enabling timely response actions that reduce vulnerability, reinforcing the resilience of communities in the face of multiple and varied environmental risks. Consider for example, households and firms' ability to self-protect against floods by purchasing and strategically placing sandbags. EWS can be applied to a broad spectrum of hazards, such as floods, storms, droughts, and landslides, with lead times that vary based on the nature of the hazard and the system's capabilities, ranging from a few minutes to several days.

Investments in EWS deliver significant risk reduction benefits. In Europe alone, it is estimated that EWS can avoid between 460 million and 2.7 billion euros of disaster-induced asset damages per year, while the figures are between US\$300 million to US\$2 billion per year in all developing countries (Hallegatte, 2012). A newer worldwide estimate suggests that potential avoided asset losses of US\$13 billion per year could be achieved through ensuring universal access to early warnings (EW) (Hallegatte et al., 2017). Avoided damage varies depending on the lead time of the warnings, as more actions can be taken with longer lead time and therefore assets protected (Table 1). From an efficiency perspective, establishing and improving EWS has proven to be very cost-effective. Several studies across different contexts and following different assumptions have reported that the cost-benefit ratios range from 1:4 (Hallegatte, 2012) to as high as 1:73 (Rai et al., 2020) or even 1:400 (Pappenberger et al., 2015).

## 2. Data

We rely on four data sources that are presented in this section.

### 2.1. World Risk Poll (WRP)

The Lloyd's Register Foundation World Risk Poll (WRP), conducted in 2019, 2021 and 2023, is a global, nationally representative survey designed to assess respondents' experiences and perceptions of risk. We mainly use the 2023 WRP survey, which includes responses from approximately 147,000 individuals across 142 countries, with most countries having a sample size of around 1,000 respondents. The survey includes questions on whether respondents experienced a disaster in the past five years and, if so, details about the most recent disaster they encountered, such as its type, whether they received an early warning, and the source of that warning. We use the WRP data in two ways: (i) through a national-level analysis that aggregates data at the country level to identify the factors influencing early warning reception at the macro scale, and (ii) through a household-level analysis that examines determinants of early warning reception at the individual level.

### 2.2. Disaster Poverty Household Survey (DPHS), Dar Es Salaam

The Disaster Poverty Household Survey (DPHS) for Dar es Salaam was conducted between 2017 and 2018 as part of an initiative funded by the Global Facility for Disaster Reduction and Recovery (GFDRR) and the Tanzania Urban Resilience Program. The DPHS is designed to examine the relationship between disaster risk—encompassing exposure, vulnerability, and recovery capacity—and poverty in urban settings. The survey specifically focuses on the impacts of urban flooding and includes information on whether households received early warnings about flooding or heavy rainfall during the preceding year. It also asks questions about the preventive actions that households undertook before the rainy season to mitigate potential disaster impacts. Our analysis uses the DPHS data to identify the factors influencing household-level reception of early warnings and to evaluate the effectiveness of early warnings in encouraging preventive actions.

### 2.3. The Bangladesh 2020 Anticipatory Action survey

The Bangladesh 2020 Anticipatory Action survey is derived from a detailed household survey conducted to evaluate the impacts of anticipatory cash transfers distributed ahead of severe flooding in Bangladesh in 2020 (Pople et al, 2021). The survey specifically collected data on whether households received early warnings about the impending flood and detailed information on the preventive actions they implemented. We use this dataset to analyze the role of early warnings in enhancing household resilience during a disaster, and to examine the determinants of early warning reception at the household level.

### 2.4. Village Potential Datasets (PODES), Indonesia

The Village Potential Datasets (PODES) is a regularly conducted survey designed to provide comprehensive data on the characteristics of villages (*desa*) across Indonesia. For this study, we use PODES data from 2019 and 2020, encompassing approximately 80,000 villages. This dataset includes information on the number of disasters experienced by each village over the previous year, the types of disasters, and the number of casualties associated with each disaster type. Additionally, it collects data on the presence of EWS in these villages. We integrate the PODES dataset with the WorldPop dataset, which provides population estimates for nearly all villages, except for 3,739 cases where data is unavailable. This dataset enables us to investigate the potential impact of EWS on reducing disaster-related casualties, a measure of disaster impacts at an individual level.

### 3. Global patterns of early warning coverage: A cross-country analysis

To examine the determinants of EW coverage at the national level, we conduct a cross-country analysis using the WRP 2023 data, aggregated at the national level.

#### 3.1. EWS coverage and reception determinants at the national level

Our findings reveal that EW coverage is unequally distributed and still far from universal (Figure 1). While in developed countries, over 60% - and often more than 80% - of the population received EW prior to the most recent disaster they experienced, coverage is notably lower in Latin America, Sub-Saharan Africa, part of Central Asia and the Middle East and North Africa. Countries in the Caribbean and East Asia, and to a lower extent South Asia are exceptions, demonstrating relatively higher EW coverage.

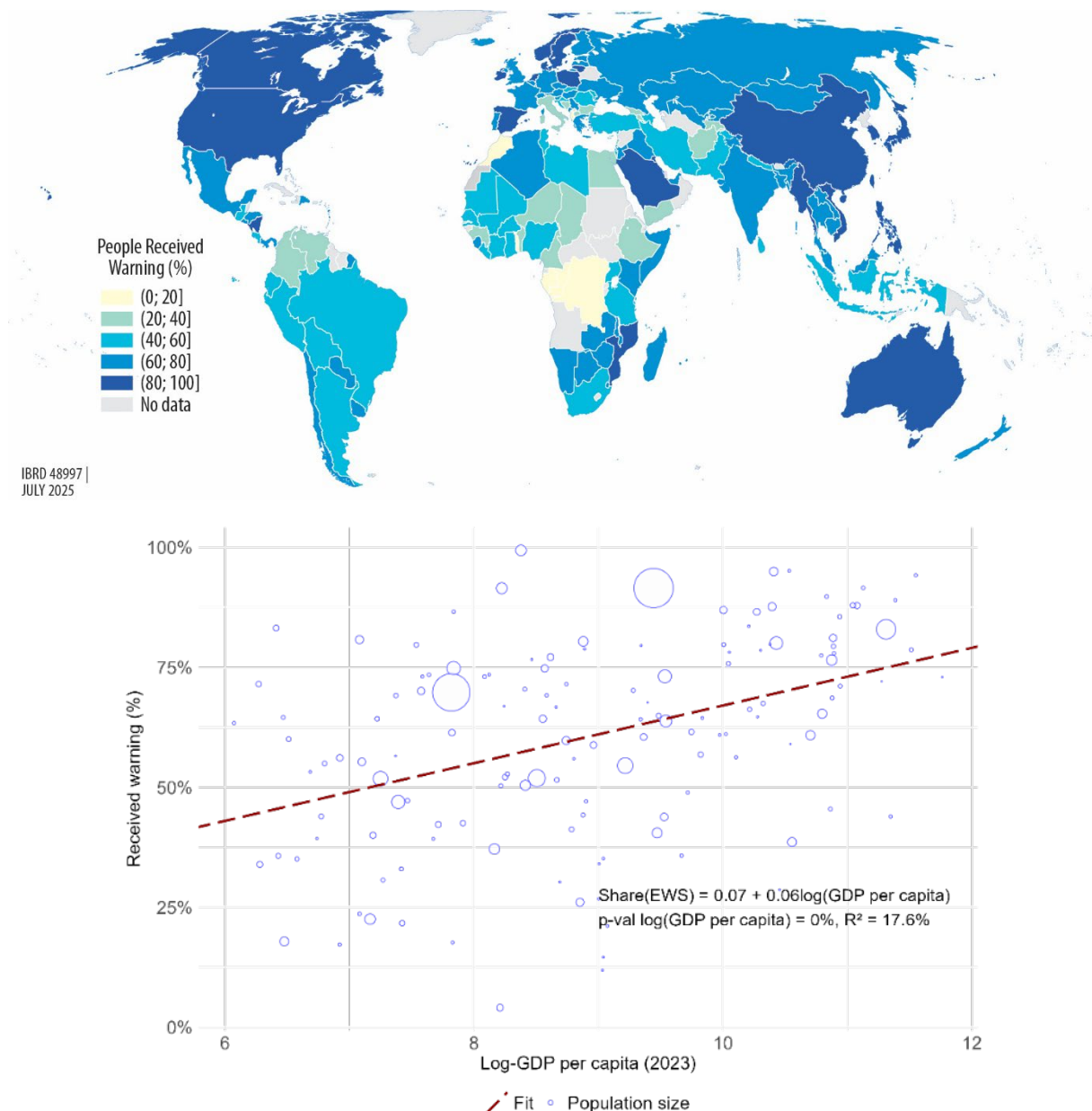


Figure 1: Overview of EW coverage. Top figure: Percentage of the population in each country who received an EW before the last disaster they experienced. Bottom figure: Association between the share

of population having received an EW before the last experienced disaster and the country's log-GDP per capita

Note: World Bank calculations based on World Risk Poll (2023)

There is a noticeable correlation between EW coverage and a country's GDP per capita: for each 1 percent increase in GDP per capita, there is an associated average 6 percentage points increase in the share of the population receiving an EW (Table 2, Model 1). Similar patterns emerge when linking EW coverage with other socio-economic indicators. EW coverage is negatively correlated with a composite multi-dimensional poverty index<sup>1</sup> and positively correlated to the human development index (Figure 4, Appendix).

However, even within similar income levels, EW coverage varies widely as can be observed from Figure 1, suggesting that per capita GDP is only one factor among others. In fact, GDP per capital alone accounts for only 17% of the variation in EWS coverage, as suggested by the adjusted R<sup>2</sup> in Table 2, Model 1.<sup>2</sup>

|                                                         | Share of people who received an Early Warning |                      |                      |                      |
|---------------------------------------------------------|-----------------------------------------------|----------------------|----------------------|----------------------|
|                                                         | Model 1                                       | Model 2              | Model 3              | Model 4              |
| Intercept                                               | 0.069<br>(0.098)                              | 0.050<br>(0.098)     | -0.228 **<br>(0.114) | -0.250 **<br>(0.118) |
| Log(GDP per capita)                                     | 0.060 ***<br>(0.010)                          | 0.060 ***<br>(0.011) | 0.066 ***<br>(0.010) | 0.066 ***<br>(0.010) |
| Number of disasters in the country in the last 10 years |                                               | 0.001 **<br>(0.000)  |                      | 0.001 **<br>(0.000)  |
| Share of people who experienced a predictable hazard    |                                               |                      | 0.327 ***<br>(0.068) | 0.329 ***<br>(0.067) |
| Number of observations                                  | 138                                           | 138                  | 138                  | 138                  |
| R <sup>2</sup>                                          | 17.6%                                         | 20.6%                | 31.3%                | 34.4%                |
| Adjusted R <sup>2</sup>                                 | 17%                                           | 19.4%                | 30.3%                | 33%                  |

*Table 2: Determinants of receiving Early Warning across countries: regression model results.*

Note: World Bank calculations based on World Risk Poll (2023)

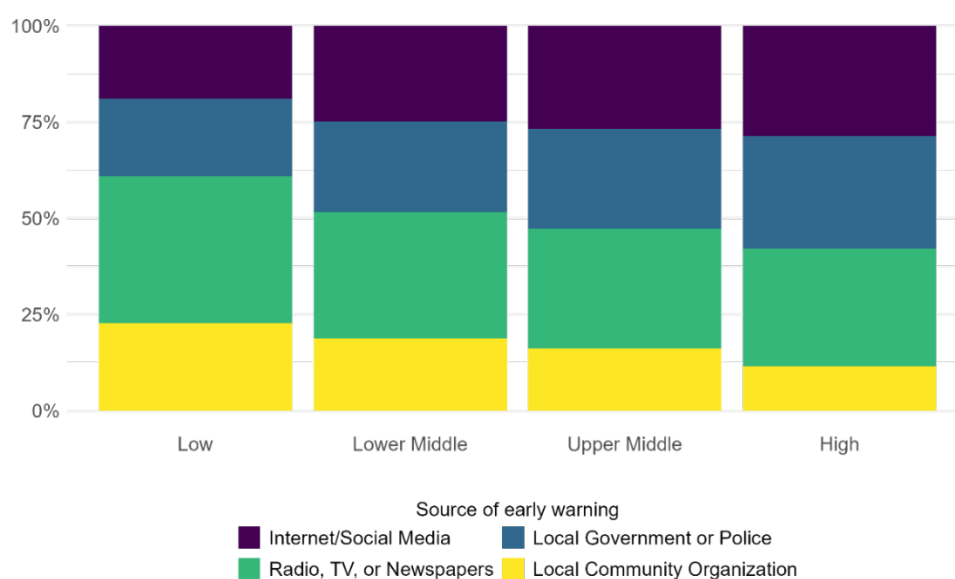
\*  $p < 0.1$ , \*\*  $p < .05$ , \*\*\*  $p < .01$

Other factors at the national level also influence EW coverage. Our analysis suggests that each additional disaster experienced by a country in the past 10 years is associated with a 0.001 percentage point increase in the population receiving early warnings (Table 2, Model 2). This suggests that

<sup>1</sup> The Multidimensional Poverty Index provides a more comprehensive understanding of poverty by considering multiple dimensions such as health, education, and living standards. Indicators used to construct the index are the following: nutrition, child mortality, years of schooling, school attendance, cooking fuel, sanitation, drinking water, electricity, housing and assets. Individuals are classified as poor if they are classified as deprived in at least one-third of the indicators. The MPI, sourced from the World Bank's Poverty indicators, is available only for 83 countries.

<sup>2</sup> This correlation (and more generally the correlations displayed in Table 2) should not be interpreted as causality as many omitted variables could be driving the relationship such as the strength of local governments, the quality of weather reports for the region or the trust of the population in public institutions.





*Figure 3: Distribution of EW information sources across income categories*

Note: World Bank calculations based on World Risk Poll (2023)

## 4. Household-level analysis of early warning reception

Reception of EW varies across households in the same country, or even in the same localities (e.g., a city or a remote village). Understanding the determinants of receiving EW before a disaster strikes at the household level is crucial, as it can help point to categories of people that are not covered and need tailored approaches to ensure that the EW reach them. To address this question, we analyze data from three distinct datasets: (i) the 2023 wave of the World Risk Poll, which offers a global perspective on the determinants of receiving EW, (ii) the DPHS survey, which focuses on urban areas in Tanzania, and (iii) the Bangladesh 2020 Anticipatory Action Survey, which examines a high-risk region in rural northern Bangladesh. These diverse datasets enable a comprehensive analysis of the factors influencing EW access across different geographical, economic, and risk contexts: the World Risk Poll 2023 provides a comprehensive overview across countries at national level, the DPHS survey provides a snapshot in an urban setting, while the Bangladesh survey gives a snapshot in a rural setting.

In these three cases, we identify the socio-economic factors at the household level that are associated with the likelihood of receiving an EW<sup>4</sup> a binary outcome variable (i.e., whether or not an EW was received). Given that the dependent variable is binary, we employ logistic regressions (logit) for the analysis.

Table 4 presents the results of the logistic regression analysis based on data from the World Risk Poll.<sup>5</sup> It reports both the estimated coefficients and the average marginal effects of each explanatory variable, allowing for the interpretation of results as the quantitative impact of each factor on the probability of receiving an early warning.

|           | Estimate              | Average marginal effect |
|-----------|-----------------------|-------------------------|
| Intercept | -0.572 ***<br>(0.123) |                         |

<sup>4</sup> We restrict the datasets to households who experienced a disaster.

<sup>5</sup> The WRP only capture the population that received an EW having experienced a disaster.



dependency ratio—representing the proportion of dependents within a household<sup>6</sup>—does not show a significant relationship with EW receipt, suggesting that household composition may not directly influence access to EW.

Education also plays a significant role. Individuals with secondary or tertiary education are 3.2 and 3.6 percentage points, respectively, more likely to receive an EW compared to those with primary education. Similarly to the income variable, this may be explained by improved access to and awareness of information of more educated households.

Urbanicity also plays a significant role in explaining the reception of EW. Households in small towns and suburbs are significantly more likely to receive an EWS compared to those in rural areas. This may be explained by the proximity of local government or police in urban areas, which facilitates the dissemination of warnings. Moreover, residents of towns may have greater access to traditional media such as radio or television, and may have a better internet connection. However, living in a large city is not associated with a statistically significant effect. This result might be attributed to the socio-economic heterogeneity of urban populations, as large cities often encompass both affluent neighborhoods with robust access to information and marginalized areas, such as informal settlements or slums, where disseminating warnings can be challenging.

We find that gender shows a modest but significant effect, with female respondents being 1 percentage point less likely to receive an EW compared to males, all else equal. This gender disparity may reflect broader societal inequalities, including differences in access to communication channels or decision-making roles, which can affect the reception of warnings. Disasters not being gender neutral and affecting women more than men through various channels is the topic of recent research (Nasreen, 2022; Reggers, 2019).

Employment status reveals mixed effects. While unemployment is not significantly associated with EW receipt compared to being employed, being out of the workforce significantly reduces the likelihood of receiving an EWS.

We further investigate the determinants of EW reception using two additional surveys: the DPHS survey, which examines an urban context in Dar es Salaam, Tanzania (Table 5), and the Bangladesh survey, which is conducted in a rural setting in Bangladesh (Table 6). Both surveys reveal notable consistencies with the WRP findings, particularly regarding the significant positive role of education in increasing the likelihood of receiving an EW. However, some discrepancies are observed. In the Bangladesh survey, the marginal effect of age on receiving an EW follows an inverted U-shape, peaking around age 50. This means the likelihood of receiving an EW increases with age up to 50, then declines thereafter. In contrast, the WRP survey shows the opposite pattern: a U-shaped relationship with a minimum around age 48—the effect decreases until 48 and then increases. The difference in patterns may be partly explained by the inclusion—or lack—of controls for mobile phone ownership. In the WRP survey, which does not control for mobile phone access, the age coefficient may be capturing the effect of mobile phone ownership itself. Since older individuals are less likely to own or use mobile phones, the marginal effect of age decreases initially—explaining the first downward phase of the U-shape. In addition, the WRP is a global survey, so interpreting this pattern precisely is more difficult due to the heterogeneity of contexts. In contrast, the Bangladesh survey controls for phone ownership, so the age effect may reflect behavior: younger adults (20s) might be less attentive to impending hazard threats, those in their 30s–40s might be more responsive, and older individuals might

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<sup>6</sup> Since the WRP does not provide information on retirees within households, we include only children in the count of dependent individuals.

levels of social integration within the local community. In the same regression analysis, we observe that the presence of an improved floor—often serving as a proxy for not residing in an informal settlement—significantly increases the likelihood of receiving an early warning. Additionally, monitoring weather conditions to predict heavy rain and flooding is a significant factor in receiving an EW, highlighting the importance of individual initiative in disaster preparedness. Furthermore, residing in a flood-prone area has a positive effect on the probability of receiving an EW, suggesting that authorities may prioritize sending warnings to individuals living in high-risk zones. In the Bangladesh survey regression, living in a protected mainland area is associated with a higher likelihood of receiving an early warning, compared to living in an unprotected mainland.<sup>7</sup> This likely reflects differences in disaster risk management: protected areas may be more exposed to flooding and therefore better covered by EWS. Moreover, receiving a dignity kit from UNFPA is also associated with a higher likelihood of receiving an early warning, possibly because it reflects a household's stronger connection to humanitarian networks and outreach efforts. It is important to note that the  $R^2$  value in this regression is notably low, indicating that the explanatory variables included in the model account for only a small proportion of the variance observed in EW reception. This suggests that additional factors, not captured by the current model, may play a significant role in influencing early warning reception.

|                                 | Estimate (logit)      | Average marginal effect |
|---------------------------------|-----------------------|-------------------------|
| Intercept                       | -0.297<br>(0.421)     |                         |
| Gender - female                 | 0.017<br>(0.148)      | 0.004<br>(0.033)        |
| Age                             | 0.030 ***<br>(0.009)  | 0.007 ***<br>(0.002)    |
| Age <sup>2</sup>                | -0.000 ***<br>(0.000) | -0.000 ***<br>(0.000)   |
| Received education              | 0.358 ***<br>(0.054)  | 0.079 ***<br>(0.012)    |
| Dependency ratio                | -0.096 **<br>(0.046)  | -0.021 **<br>(0.010)    |
| Own a mobile phone              | 0.108 *<br>(0.058)    | 0.024 *<br>(0.013)      |
| Protected mainland              | 0.150 **<br>(0.064)   | 0.033 **<br>(0.014)     |
| Received dignity kit from UNFPA | 0.294 ***<br>(0.077)  | 0.063 ***<br>(0.016)    |
| Treatment - cash transfer       | 0.063<br>(0.056)      | 0.014<br>(0.013)        |
| Union FE                        |                       | YES                     |
| Nb obs                          |                       | 9112                    |
| McFadden pseudo- $R^2$          |                       | 3.5%                    |

<sup>7</sup> Protected mainland areas are those located inside flood embankments, whereas unprotected mainland areas are those located outside the embankments.

|                | Estimate (logit) | Average marginal effect |
|----------------|------------------|-------------------------|
| Log-likelihood |                  | -5774.97                |
| AIC            |                  | 11673.94                |

Table 6: Factor influencing receipt of EW at the household level: logit result

Note: World Bank calculations based on Bangladesh 2020 Anticipatory Action survey

\*  $p < 0.1$ , \*\*  $p < .05$ , \*\*\*  $p < .01$ .

## 5. The benefits of early warnings on household resilience and safety

### 5.1. EWS strengthen households' (perceived) resilience

The World Risk Poll estimates an aggregated resilience index, which ranges from 0 to 100, based on factors that influence one's resilience at individual (e.g., agency and self-efficacy<sup>8</sup>), household (e.g., financial ability to cover disaster impacts), community (e.g., satisfaction of existing infrastructure in the community), and societal (e.g., perception of trust toward public servants) levels. Therefore, the resilience index reflects both the subjective experiences and objective conditions influencing resilience. To understand whether access to EW is positively associated with the resilience index,<sup>9</sup> we perform a regression analysis of the World Risk Poll's Resilience Index on key predictors, including disaster experience, receipt of EW, and a range of socio-economic variables. The regression reveals that while experiencing a disaster has a significant and negative impact on the resilience index (reduction by on average 1.71), the receipt of an early warning increases the resilience index by 4.3 on average, which more than compensates for the experience of a disaster.<sup>10</sup>

|                         | Resilience index      |
|-------------------------|-----------------------|
| Intercept               | 26.602 ***<br>(0.463) |
| Disaster experienced    | -1.714 ***<br>(0.119) |
| Received warning        | 4.302 ***<br>(0.141)  |
| Socio-economic controls | YES                   |
| Country FE              | YES                   |
| Nb obs                  | 143808                |
| R <sup>2</sup>          | 38.8%                 |
| Adjusted R <sup>2</sup> | 38.7%                 |

Table 7: Factor influencing the resilience index at the household level: OLS results

<sup>8</sup> Agency/self-efficacy refer to the belief in one's ability to take protective actions in the face of a potential disaster, as captured by the question "If a disaster were to occur near you in the future, do you think there is anything you could do to protect yourself or your family from its impact?"

<sup>9</sup> Note that the calculation of the composite resilience index does not include access to EW, hence the analysis remains valid.

<sup>10</sup> To formally assess whether the positive effect of receiving an early warning more than offsets the negative effect of disaster experience, we conduct a linear hypothesis test of the sum of the two coefficients. The test rejects the null hypothesis that their combined effect is zero, indicating that receiving a warning more than compensates for the adverse impact of experiencing a disaster on household resilience.

Note: World Bank calculations based on World Risk Poll (2023)

\*  $p < 0.1$ , \*\*  $p < .05$ , \*\*\*  $p < .01$ .

## 5.2. EWS increases the likelihood of taking preventive actions

The DPHS survey and the Bangladesh survey investigate respondents' engagement in preventive measures prior to floods. The DPHS survey emphasizes actions taken before the onset of the rainy season,<sup>11</sup> while the Bangladesh survey focuses on measures implemented in preparation for the most recent flood. We examine the correlation between receiving a warning and the adoption of preventive measures through the use of logistic regressions, controlling for socio-economic variables and other relevant factors. For the DPHS survey, these include whether the household monitors weather forecasts to predict floods or heavy rainfall, and whether the dwelling is located in a flood-prone area. For the Bangladesh survey, variables capturing the extent of the flooding are also considered. The findings from the logistic regression analyses are summarized in Table 8 and Table 9.

The findings from the DPHS survey indicate a significant positive association between receiving an EW and adopting various preventive measures. These measures include maintaining house surroundings clean to prevent waste clogging drains, ensuring proper drainage, participating in prevention activities organized at the neighborhood level, digging a ditch around the house, building a wall, and placing sandbags around the house. Note that we acknowledge the discrepancy in the temporality of the two variables under consideration. The question regarding EW reception pertains to whether respondents received an early warning prior to or during heavy rainfall within the past year. In contrast, the questions concerning preventive actions refer to whether households generally implement these measures as part of their usual practices. The positive coefficient associated with the EW variable may reflect increased household awareness of the importance of taking preventive action as a result of receiving an EW. However, it could also be attributable to other unobserved variables.

Similarly, the results from the Bangladesh survey reveal that individuals who received an EW are more likely to undertake critical protective actions such as evacuating household members, purchasing food, evacuating livestock, reinforcing roofs and walls, and warning others. These results underline the crucial role of EW in fostering proactive behavior and improving households' resilience to floods and heavy rains.

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<sup>11</sup> Note that we are aware that the temporality of both variables are not the same. The question about EW reception is whether they received an early warning prior or during heavy rain in the last year, whereas the questions about the preventive action is whether or not they usually put these actions in place. The positive coefficient associated to the variable EW can therefore correspond to the fact that household's awareness of the importance to take action is raised thanks to the early warning, but it can also be due to the fact that they have been hit by a disaster in the past years, and we cannot control for this.

| Average marginal effect         |                       |                         |                            |                       |                       |                        |                       |
|---------------------------------|-----------------------|-------------------------|----------------------------|-----------------------|-----------------------|------------------------|-----------------------|
|                                 | Any action            | Protect valuable assets | Evacuate household members | Purchase food         | Evacuate livestock    | Protect roof and walls | Warn others           |
| Received an EW                  | 0.330 ***<br>(0.007)  | 0.021<br>(2.201)        | 0.189 ***<br>(0.010)       | 0.323 ***<br>(0.009)  | 0.139 ***<br>(0.009)  | 0.074 ***<br>(0.007)   | 0.067 ***<br>(0.007)  |
| Own a mobile phone              | 0.020<br>(0.012)      | 0.000<br>(0.001)        | 0.026 **<br>(0.012)        | 0.023 *<br>(0.012)    | 0.002<br>(0.010)      | 0.001<br>(0.006)       | 0.015 ***<br>(0.005)  |
| Brick house                     | -0.558 ***<br>(0.006) | -0.001 ***<br>(0.000)   | -0.282 ***<br>(0.006)      | -0.389 ***<br>(0.005) | -0.163 ***<br>(0.004) | -0.050 ***<br>(0.002)  | -0.050 ***<br>(0.002) |
| Protected mainland              | -0.007<br>(0.014)     | -0.000<br>(0.001)       | -0.047 ***<br>(0.013)      | 0.034 **<br>(0.014)   | 0.009<br>(0.011)      | -0.033 ***<br>(0.007)  | -0.001<br>(0.006)     |
| Received dignity kit from UNFPA | 0.038 **<br>(0.016)   | -0.001 ***<br>(0.000)   | -0.025 *<br>(0.015)        | 0.046 ***<br>(0.015)  | -0.003<br>(0.012)     | -0.024 ***<br>(0.006)  | -0.015 **<br>(0.006)  |
| Treatment - cash transfer       | 0.028 **<br>(0.012)   | 0.001<br>(0.001)        | 0.012<br>(0.011)           | 0.010<br>(0.012)      | 0.025 ***<br>(0.009)  | -0.008<br>(0.006)      | -0.018 ***<br>(0.006) |
| Maximum extent of flooding      | -0.019<br>(0.040)     | -0.005<br>(0.004)       | 0.007<br>(0.039)           | -0.033<br>(0.040)     | 0.019<br>(0.032)      | 0.001<br>(0.019)       | 0.039 **<br>(0.018)   |
| Flood duration                  | 0.001 ***<br>(0.000)  | 0.000<br>(0.000)        | -0.001 *<br>(0.000)        | 0.002 ***<br>(0.000)  | 0.001<br>(0.000)      | -0.000<br>(0.000)      | -0.000<br>(0.000)     |
| Socio-economic controls         | YES                   | YES                     | YES                        | YES                   | YES                   | YES                    | YES                   |
| Union FE                        | YES                   | YES                     | YES                        | YES                   | YES                   | YES                    | YES                   |
| Nb obs                          | 9030                  | 9030                    | 9030                       | 9030                  | 9030                  | 9030                   | 9030                  |
| McFadden pseudo-R <sup>2</sup>  | 14.2%                 | 31.7%                   | 7.6%                       | 15%                   | 8.2%                  | 17.8%                  | 11.4%                 |
| Log-likelihood                  | -5318                 | -57.95                  | -4963.32                   | -5127.64              | -3690.47              | -1472.48               | -1591.36              |
| AIC                             | 10766                 | 245.9                   | 10056.63                   | 10385.28              | 7510.94               | 3074.97                | 3312.72               |

Table 9: Have You Taken the Following Actions before the flood? Logit results

### 5.3. EWS reduce the number of casualties

To examine the impact of EWS on casualty rates resulting from various types of disasters, we use the Indonesia PODES datasets for the years 2019 and 2020. These datasets capture, for each admin-4 area,<sup>12</sup> the annual disaster occurrence and associated casualty counts disaggregated by disaster type, with the casualty data for year n-1 and year n from January to April (2019) or May (2020). For instance, the 2020 PODES dataset includes casualty data for the entire year 2019 and for the year 2020 from January to May. Each dataset also includes village-level information regarding the presence of an EWS. Therefore, to assess the potential protective effect of EWS on reducing casualties, we use the 2019 PODES dataset to determine EWS presence in villages and the 2020 dataset to obtain the disaster and casualty counts specific to 2019.

We first analyze the casualty rate, i.e., the average casualty per event per 1,000 people, at the admin-4 level in Indonesia in 2019 (Table 10). All hydrometeorological events in this analysis (flood, flash flood, tidal wave and cyclone) show lower values of casualties per event per 1,000 people for villages with EWS compared to those without EWS. For example, the casualties per event for flood is 0.096 in villages without EWS, while it is 0.005 in villages with EWS, or 19 times lower than in villages without EWS. The general trend is similar for the other hazards including flashflood and cyclones, where the casualties per event per capita were lower in subdistricts with EWS.

| Type of disaster          | Subdistricts without EWS                      |                              | Subdistricts with EWS                         |                              |
|---------------------------|-----------------------------------------------|------------------------------|-----------------------------------------------|------------------------------|
|                           | Disaster Casualties per event per 1000 people | Number of districts affected | Disaster Casualties per event per 1000 people | Number of districts affected |
| Flash flood               | 0.058                                         | 649                          | 0.006                                         | 64                           |
| Flood                     | 0.096                                         | 9,459                        | 0.005                                         | 743                          |
| Ocean tidal wave          | 0.035                                         | 865                          | 0.005                                         | 164                          |
| Whirlwind/tornado/typhoon | 0.054                                         | 2,597                        | 0.004                                         | 260                          |

*Table 10: Disaster casualties per event per 1000 people, in the year 2019*

Note: World Bank calculations based on PODES (2019 & 2020)

Further, we conduct multivariate regressions on the casualty rate. While this analysis does not establish a causal effect of EWS implementation, controlling for confounding factors allows us to approach a more accurate estimation of its impact. We restrict the analysis to predictable hazards: floods, fast floods, tidal waves and cyclones. We use negative binomial regression to model the number of casualties per event per 1,000 people.<sup>13</sup> Table 11 displays results of the regressions at the

<sup>12</sup> Admin-4 unit in Indonesia refers to *desa* or *kelurahan*, i.e., rural or urban villages.

<sup>13</sup> We first use a Poisson regression to model the number of casualties per event per 1,000 people. Poisson regression is suitable for count data, particularly for rare events such as casualties from disasters. However, one of the key assumptions of the Poisson distribution is mean-variance equality, conditional on explanatory variables. To check this assumption, we conduct the dispersion test proposed by Cameron & Trivedi (1990), and find evidence for overdispersion for all the regressions we model, indicating that the variance exceeds the mean. Therefore, we use a Negative Binomial (NB) regression as an alternative. The NB model, like the Poisson model, assumes a discrete probability distribution, making it suitable for our analysis. However, it incorporates an additional dispersion parameter that allows the variance to increase at a faster rate than the mean, addressing the overdispersion observed in the data.

village and type of disaster level,<sup>14</sup> and the incidence rate ratio (IRR).<sup>15</sup>

|                                                                 | Estimate (s.e.)     | IRR<br>[Confidence interval] |
|-----------------------------------------------------------------|---------------------|------------------------------|
| Intercept                                                       | -3.10 ***<br>(0.19) | 0.04 ***<br>[0.03; 0.07]     |
| Presence EWS                                                    | -1.62 ***<br>(0.48) | 0.20 ***<br>[0.06; 0.71]     |
| Number medical facilities/100000 hab                            | 0.05 **<br>(0.02)   | 1.05 **<br>[1.01; 1.15]      |
| Presence safety equipment                                       | -0.14<br>(0.59)     | 0.87<br>[0.24; 3.98]         |
| Presence disaster evacuation routes                             | -0.80 *<br>(0.47)   | 0.45 *<br>[0.16; 1.49]       |
| Construction, maintenance of canals, etc.                       | -0.59 **<br>(0.25)  | 0.55 **<br>[0.32; 0.99]      |
| Road accessible except certain times/only during the dry season | 1.66 ***<br>(0.44)  | 5.28 ***<br>[2.05; 15.37]    |
| Road never accessible                                           | -0.29<br>(0.82)     | 0.75<br>[0.15; 5.86]         |
| No road (traffic by air or water only)                          | 0.39<br>(0.97)      | 1.47<br>[0.25; 19.65]        |
| Weak or no internet signal                                      | 0.14<br>(0.41)      | 1.15<br>[0.42; 3.03]         |
| Weak or no cellphone signal                                     | 0.35<br>(0.34)      | 1.42<br>[0.67; 3.24]         |
| Disaster type: Flash flood                                      | 0.34<br>(0.54)      | 1.40<br>[0.48; 5.28]         |
| Disaster type: Ocean tidal wave                                 | -0.07<br>(0.48)     | 0.93<br>[0.36; 2.84]         |
| Disaster type : Whirlwing/tornado/typhoone                      | 0.07<br>(0.30)      | 1.07<br>[0.59; 2.05]         |
| Nb obs                                                          | 14801               |                              |

*Table 11: Determinants of the rate of casualties per event and per 1000 people: NB regression results*

Note: World Bank calculations based on PODES

\*  $p < 0.1$ , \*\*  $p < .05$ , \*\*\*  $p < .01$ .

<sup>14</sup> Consequently, if a village experienced multiple types of disasters, it may appear multiple times in the dataset.

<sup>15</sup> In a negative binomial regression, the incidence rate ratio (IRR) represents the expected change in the count outcome for a one-unit increase in the predictor, expressed as a multiplicative factor relative to the baseline rate. An IRR greater than 1 indicates a positive association, while an IRR less than 1 indicates a negative association.



The analysis reveals that the presence of an EWS in an admin-4 unit significantly reduces the casualty rate per event and per 1,000 inhabitants. The coefficient of -1.62 is highly significant and corresponds to an IRR of 0.20. This indicates that having an EWS divides the casualty rate by a factor of 5.

We also uncover other interesting findings regarding the determinants of the casualty rate. The presence of a disaster evacuation route is associated with a reduction in the number of casualties, probably because it enables more efficient evacuation during disasters. Similarly, admin-4 units with roads that are accessible only during certain times or in the dry season experience significantly higher casualty rates compared to those with roads accessible year-round. This might be because emergency measures and medical aid face greater difficulties in reaching these areas during a disaster. However, no significant impact on casualties is observed for villages with no road access (access only by air or water) or having roads that are always inaccessible. Interestingly, the number of medical facilities in a village is positively associated with the casualty rate, a result that may initially appear counterintuitive. This could be due to (i) admin-4 units with medical facilities serving as regional hubs, where deaths of patients from surrounding areas are recorded or (ii) reporting bias, as admin-4 units with better health infrastructure likely document fatalities more comprehensively. Finally, the construction and maintenance of canals are expectedly associated with a significant reduction in the number of casualties.

## 6. Conclusion

This study leverages several surveys conducted at international, national, and local scales, including in both urban and rural settings to provide a rich and multi-faceted analysis of EWS coverage and sources worldwide, factors influencing households' early warning reception, and the associated benefits in enhancing disaster resilience and reducing disaster-related impacts. It plugs a gap in the literature which tended to either have a global focus about the benefits of EWS or be ultra-local in nature, with question marks surrounding how transferable their findings were across circumstances. Our approach asks reasonably similar questions over various geographic areas, and its added value resides in presenting these in a unified framework which is underpinned by a bottom-up, people centered lens to understand the coverage, inclusivity, receipt determinants and benefits of early warning systems and early warnings.

Our findings highlight significant disparities in EWS coverage across countries, with higher-income nations demonstrating greater accessibility compared to lower-income regions. Socioeconomic and demographic factors, including age, gender, income level, education, urbanicity, and employment status play a crucial role in determining whether individuals receive early warnings. The specific Tanzania and Bangladesh deep-dives also highlight asset ownership and dwelling location and quality as additional determinants.

The results underscore the critical role of EWS in improving disaster resilience. Households that receive timely warnings are more likely to take preventive measures, such as evacuating, securing assets, and reinforcing their homes, ultimately reducing the impact of disasters. Moreover, our analysis of casualty data suggests that the presence of EWS significantly lowers the number of fatalities and injuries during disaster events. As shown in the Indonesia deep-dive, the presence of EWS in subdistricts corresponds to lower casualty rates by a factor as high as five.

From a policy perspective, these findings highlight the urgent need to enhance the accessibility and effectiveness of EWS, particularly in low-income and disaster-prone regions, and for disadvantaged and vulnerable segments of the population. Strategies to expand coverage should focus on strengthening, expanding, and diversifying communication channels, since the ways people receive warnings vary – from internet and social media, local officials, local organizations, and even radios. The

# Costs of Health Care Associated Infections from Inadequate Water and Sanitation in Health Care Facilities in Eastern and Southern Africa

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## Abstract

In Sub-Saharan Africa, health care facilities face critical challenges in water supply, sanitation, and hygiene services; health care waste management; and environmental cleanliness. With coverage below 50 percent, these deficiencies pose significant health risks to patients and health care workers, contributing to health care–associated infections. Meta-analyses and individual studies estimate rates of health care–associated infections in Sub-Saharan Africa at between 13 and 30 percent of hospital admissions, impacting patients, families, and health care providers. Rising antimicrobial resistance further exacerbates health outcomes and costs. In Eastern and Southern Africa, an estimated 3.1 million health care–associated infections in 2022 incurred

over 320,000 excess deaths, costing at least US\$6 billion, or 1.14 percent of combined gross domestic product in 2022. Investing in comprehensive water supply, sanitation, and hygiene and health care waste management can yield substantial benefits, with a benefit-cost ratio of 5.8 for all economic costs. Beyond preventing health care–associated infections, improved cleanliness and infrastructure are crucial for patient satisfaction, impacting future health care–seeking behavior and health care worker job satisfaction. Sub-Saharan African countries should prioritize infrastructure investment, budget allocation, staffing, and behavioral improvements to enhance the quality of health care and mitigate these pressing challenges.

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## 1. Introduction

### Importance and causes of healthcare associated infection

Healthcare associated infections (HAI) remain a global health challenge. An HAI is a condition resulting from an infection acquired while a patient is present in a health care setting (Horan et al, 2008).<sup>1</sup> HAIs impact patients and their families by causing illness, prolonged hospital stay, potential disability, excess costs and sometimes death. HAIs help drive the challenges associated with antimicrobial resistance (AMR) and as a result are more costly to treat and impact the patient's health more (Serra-Burriel et al, 2020). Rates of HAI are elevated in Sub-Saharan Africa due to poor infection control practices, inappropriate use of limited resources, under-staffing of healthcare facilities and overcrowding of hospitals (Irek et al, 2018; Ogunsola et al; 2020). Pooled estimates from Abubakar et al (2022) estimate 12.76% (95% confidence interval 10.30 – 15.23) of inpatients are infected with an HAI. However, single-hospital studies from SSA show significantly higher rates, especially for surgical site infections.

HAIs have multiple causes. Khan et al (2017) identify three major reservoirs and transmission routes: (1) endogenous native microflora of patient leading to cross-infection from one site to another; (2) direct patient-to-patient or staff-to-patient pathogen transmission (hands, saliva, fecal matter); and (3) other environmental sources (water, food, equipment, surfaces, and waste). Watson et al (2019) map the transmission pathways for different types of HAIs stemming from inadequate water, sanitation, hygiene, and HCWM services, shown in Figure 1. Health care waste, especially contaminated surgical waste, often acts as a reservoir for pathogenic virulent microorganisms, and it is suggested that 20%–25% of the waste produced by health care outlets is of a toxic nature and considered to have high potential to cause HAIs (Khan et al, 2017). HAIs can also be due to implants and prostheses, most commonly including central line-associated bloodstream infections (CLABSIs), catheter-associated UTIs, and ventilator-associated pneumonia (VAP) (United States CDC, 2023).

**Figure 1. Conceptual framework visualizing the relationship between water, sanitation, and hygiene (WASH) and HAIs.**

## Disease burden associated with healthcare associated infections

HAIs lead to massive preventable health burden. It is estimated that approximately 20% of all-cause global deaths are due to sepsis: around 11 million potentially avoidable deaths per year (WHO, 2020; Rudd et al, 2020). In SSA, sepsis accounts for 3% of maternal complications and 10% of maternal deaths (Bailey et al, 2017). There are an estimated 2.6 million cases of neonatal sepsis per year leading to 250,000 deaths (Seale et al 2014). In East Africa the neonatal sepsis rate is estimated at 30% of live births (Abate et al, 2020). According to Zaidi et al (2005), more than half of all infants in neonatal units in developing countries acquire an HAI, with a fatality rate ranging between 4%–56%.

Antimicrobial resistance is a major cause for concern, given the waning efficacy of frontline drugs to treat bacterial and viral infections. AMR is the ability of a microorganism (such as bacteria, viruses, fungus, and some parasites) to stop an antimicrobial (such as antibiotics, antivirals, antifungals and antimalarials) from working against it. As a result, standard treatments become ineffective, infections persist and worsen, and may spread to other patients and to healthcare workers. Poor infection control and inadequate sanitary conditions contribute to the spread of AMR.

WASH, HCWM and environmental cleaning services play a key role in stemming the tide of HAIs. Monegro et al (2023) state that universal standard (infection control) measures, such as handwashing with soap and water or using alcohol-based disinfectant before and after each patient visit, are vital in reducing rates of transmission of multi-drug resistant pathogens. Similarly, the WHO practical guide on hospital acquired infections advises three main prevention strategies: patient risk classification, patient-to-patient hygiene measures, and environmental hygiene measures (Ducel et al, 2002). WHO's World Alliance for Patient Safety advocates that effective hand hygiene is the single most important practice to prevent and control HAIs, which form colonies with multi-drug resistant microbes (Pittet et al, 2006).

## The economics of healthcare associated infections

In implementing measures to control HAIs in general, and AMR in particular, economic considerations are vital to consider in making the best use of limited healthcare resources. Several types of economic study provide different perspectives on the control of HAIs, including cost-of-illness, cost of prevention, cost-benefit analysis, cost-utility analysis, and cost consequence analysis. A rich literature exists on the health and economic consequences of HAIs, although this is mainly in OECD countries (Serra-Burriel et al, 2020).

**Cost-of-illness studies** estimate the medical costs and economic losses due to HAIs. For SSA, Epeh and Fenny (2022) report results for 13 studies estimating additional medical costs and average length of stay (ALOS) due to HAIs, finding significant heterogeneity across studies. In Ethiopia, Gidey et al (2023) compare the cost of treating HAI versus non-HAI patients and find an additional cost of US\$58 per HAI (US\$6.0 for procedures and investigations, US\$4.7 for hospital bed and US\$47.2 for drugs) and additional ALOS of 8.3 days. In Ghana, Otioku et al (2023) find that having an HAI increases drug cost by 9% and increased laboratory cost by 85%, and ALOS by 3 days. In Ethiopia, pediatric admissions with HAI have 5 days longer ALOS than without HAI (Sahiledengle et al, 2020). In Tanzania, postoperative hospital stay is 7.6 days longer for patients with HAI (Eriksen et al, 2003). In Rwanda, HAI increase ALOS by 3.6 days (Sutherland et al, 2019).

Antimicrobial resistance increases the costs of medical care, through the need for more procedures, drugs and longer stay. In Ghana, Otioku et al (2023) found that an AMR infection increased drug cost by 30%, increased laboratory cost by 42%, and ALOS by 4.9 days. From 20 studies (19 OECD countries and 1 China), multi-drug resistant HAIs cost an additional 33%, with 27% increase in ALOS (Serra-Burriel et al, 2020).

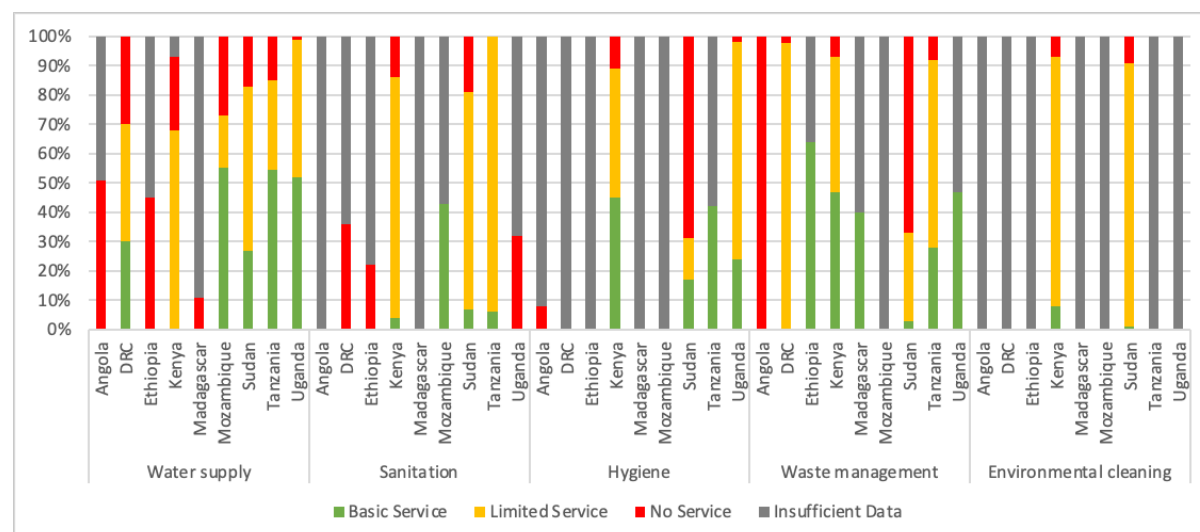
**At the macroeconomic level**, the World Bank concludes that the economic costs of AMR may be within the same order of magnitude as those of the major 2008 global financial crisis, or a loss of global GDP of 0.14% every year, with more pronounced negative impacts in low-income countries (World Bank, 2017). Their estimates include AMR in both animals and humans. Dadgostar (2019) reports that about 10 million people will die across the world annually by 2050 if strong and effective action against AMR is not taken and could cost health systems from US\$300 billion to more than US\$1 trillion annually by 2050 worldwide. OECD (2018) estimates that between 2015 and 2050, up to US\$3.5 billion (at Purchasing Power Parity) is expected to be spent yearly on AMR-related complications across 33 OECD and EU countries. This corresponds to 10% of health care costs caused by communicable diseases.

**The costs of interventions** to improve basic WASH and HCWM services in healthcare facilities was estimated for 46 low-income countries (Chaitkin et al, 2022), finding that the cost per capita needed is US\$0.30 for capital cost and starting at US\$0.10 for recurrent cost going up to US\$0.51 per year at full coverage (Chaitkin et al, 2022). A review on the costs of WASH in healthcare facilities in low- and middle-income countries found 36 studies, with piecemeal information for different service types and levels but limited evidence for giving a picture of the resource needs on the continent (Anderson et al, 2021).

## Study focus on Eastern and Southern Africa

In SSA, there is a lack of routine monitoring on WASH, HCWM and environmental cleaning in healthcare facilities, although it has been improving since the WHO/UNICEF Joint Monitoring Programme (JMP) started compiling statistics in the SDG era. Figure 3 shows the latest data available in the nine countries from Eastern and Southern Africa included in this study. Data on environmental cleaning are available for only 2 of the 9 countries, and there are major data gaps for basic sanitation and basic hygiene. Lack of data would suggest that these basic conditions for patient and health worker safety are largely absent. To provide a better picture of coverage in healthcare facilities, these data compiled by the JMP can be supplemented with other data available from the study countries, although attention is needed on quality and national representation of additional data. For example, in Tanzania between 2005 and 2010, it was found that only 24% of health facility delivery rooms were “WATSAN-safe” (Benova et al, 2014).

**Figure 3. Coverage of WASH, HCWM and environmental cleaning in healthcare facilities in 9 Eastern and Southern African in 2021**



Source: Global progress report on water, sanitation and hygiene in health care facilities: fundamentals first. Geneva: World Health Organization; 2020. License: CC BY-NC-SA 3.0 IGO.

The SSA region is disproportionately affected by AMR, with rates well over 50% for many frontline drugs (Murray et al, 2019), in part owing to the high burden of infectious diseases, poor regulation of antimicrobial use, and a lack of alternatives to ineffective antimicrobials (Kariuki et al, 2022). For neonatal sepsis, 31% of mortalities (690,000) reported were plausibly attributed to antimicrobial-resistant infections (Li et al, 2020).

In conclusion, while evidence is quite favorable for the return on investing in WASH, HCWM and environmental cleaning, the evidence for SSA is weak. This study aims to fill several gaps in our understanding of the damage costs of inadequate WASH, HCWM and environmental cleaning in healthcare facilities. The study takes a more comprehensive perspective on the health economic impacts of poor WASH, HCWM and environmental cleaning in healthcare facilities than other studies in the literature, including the value of patient productivity and avoided mortality in the estimates. Second, the study fills a gap in the literature and provides a multi-country assessment of the cost of HAIs in several African countries.

## 2. Methods

### Overall study scope

Using a standard cost-of-illness methodology, a quantitative model was constructed to combine several variables to estimate the economic costs to society of HAIs. Nine countries of Eastern and Southern Africa were selected based on their contribution to lack of WASH, HCWM and environmental cleaning in healthcare facilities in this region: Angola, DRC, Ethiopia, Kenya, Madagascar, Mozambique, Sudan, Tanzania, and Uganda.

Three categories of cost were summed up to estimate the cost-of-illness (or damage costs) of HAIs: medical costs, productivity costs and mortality costs. To provide relevant information for healthcare decision makers, the proportion of HAIs that could be prevented through comprehensive WASH, HCWM and environmental cleaning measures were assessed and compared with the approximate costs of these control measures, to generate a benefit-cost ratio.

All estimates are made in United States dollars using average exchanges rates in 2022 and costs are presented in 2022 prices. Given the uncertainty of the parameters within the model, sensitivity analysis is conducted on selected parameters.

Other impacts of poor WASH, HCWM and environmental cleaning in healthcare facilities are less well quantified in the literature. Several impacts relating to quality of care and patient satisfaction are described by drawing on global and African literature. The combined quantitative and qualitative analysis therefore provides an overall picture of the consequences of inadequate WASH, HCWM, and environmental cleaning services in healthcare facilities.

### Economic consequences of healthcare-associated infections

Several consequences of HAIs have an associated cost, as reviewed by Morel et al (2020), Shrestha et al (2018) and McGowan (2001). While these cited studies focus on AMR HAIs, they are relevant for HAIs more broadly. Morel et al (2020) develop a comprehensive list of direct and indirect costs. Shrestha et al (2018) further categorize the costs from the perspective of patients, the health system, and society at large. Due to data and time limitations, the present study does not estimate all the costs identified as being relevant. The costs included are micro-economic costs based on the estimated number of HAIs, calculated as the proportion of inpatients being infected with an HAI multiplied by the total inpatient admissions in each country in one year, and the associated patient and healthcare costs (see Table 1). Further details on data sources and values used are provided in Annex 1.

Healthcare costs are estimated based on the number of HAIs annually multiplied by the additional cost of treating a patient with an HAI (drug and laboratory costs and longer average length of stay (ALOS)), separately for AMR and non-AMR HAIs. Costs are estimated differently for patients admitted at health clinic (primary), district hospital (secondary) and higher hospital (tertiary) levels, using unit costs per inpatient bed and drug and laboratory costs at each level (see Table 1).

Productivity losses are estimated based on the number of annual HAIs multiplied by the average number of days of additional time loss due to HAI multiplied by the daily value of time. The time losses of both patient and caregiver time are included. The daily value of time is conservatively estimated using the value-added per agricultural worker.



**Table 1. Variables, data needs and sources for economic assessment of HAIs**

| Cost variable                                                  | Data needs                                          | Data value or range across countries (range for sensitivity analysis)                                                | Data source                                                     |
|----------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Healthcare cost-of-illness</b>                              |                                                     |                                                                                                                      |                                                                 |
| Annual healthcare associated infections (HAIs)                 | Average % of admissions infected with HAI           | 12.76% (6.4% - 25.6%) used in 8 countries                                                                            | Abubakar et al (2022)                                           |
|                                                                |                                                     | 17.0% (8.5% - 34.0%) used in Ethiopia                                                                                | Alemu et al (2020)                                              |
|                                                                | Proportion of HAIs that are AMR                     | 50% (25% to 70%) <sup>2</sup>                                                                                        | Kariuki et al (2022), Murray et al (2022)                       |
|                                                                | No. of annual hospital admissions                   | 2.5 – 11.2 admissions per 100 population <sup>3</sup>                                                                | Ministry of Health                                              |
| Additional cost of treating a patient with an HAI <sup>4</sup> | Average hospital cost per day ('hotel' cost)        | I: US\$5.98-US\$55.6<br>II: US\$9.6-US\$54.06<br>III: US\$17.63-US\$206.4 <sup>5</sup>                               | IHME (2014, 2015)                                               |
|                                                                | Cost of drugs and laboratory tests per HAI (no AMR) | I: US\$11.8 to US\$49.0 <sup>6</sup><br>II: US\$28.2-US\$117.6 <sup>7</sup><br>III: US\$129.5-US\$539.5 <sup>8</sup> | Gidey et al (2023)<br>Aerts et al (2022)<br>Bocoum et al (2019) |
|                                                                | Cost of drugs and laboratory tests per HAI with AMR | Double the costs of HAI (above) for each level of care                                                               | Assumption                                                      |
|                                                                | Additional length of stay HAI (no AMR)              | 5 days (3 to 10 days) <sup>9</sup>                                                                                   | Fenny et al (2020)                                              |
|                                                                | Additional length of stay HAI with AMR              | 10 days (6 to 20 days) <sup>10</sup>                                                                                 | Otieku et al (2023)                                             |
| <b>Productivity cost-of-illness</b>                            |                                                     |                                                                                                                      |                                                                 |
| Annual HAIs                                                    | Same as Healthcare cost                             |                                                                                                                      |                                                                 |
| Additional days for recovery after HAI                         | Additional recovery time after having HAI           | 2 days                                                                                                               | Assumption                                                      |
| Loss in value-added of workers                                 | Productivity loss (agricultural value-added)        | US\$2.1 to US\$22.0 per day (US\$9 to US\$284 per day in industry) <sup>11</sup>                                     | World Bank statistics                                           |
|                                                                | Proportion of HAI patients that are adults          | 60%                                                                                                                  | Assumption                                                      |
|                                                                | Labor force participation rate                      | 28.3% to 83.6% (female)<br>50.8% to 89.2% (male) <sup>12</sup>                                                       | ILOSTAT (2023)                                                  |
|                                                                | Proportion of HAIs suffered by women                | 60%                                                                                                                  | Assumption                                                      |
| Loss of time spent in non-productive activities                | Average daily value (opportunity cost)              | 30% of the agricultural value-added                                                                                  | Assumption                                                      |
| <b>Mortality cost-of-illness</b>                               |                                                     |                                                                                                                      |                                                                 |
| Annual HAIs                                                    | Same as Healthcare cost                             |                                                                                                                      |                                                                 |
| Case fatality rate                                             | Case fatality rate HAI (no AMR)                     | 6.9% (4% to 15%) <sup>12</sup>                                                                                       | Gidey et al (2023) and 6 other studies – see Table 9            |
|                                                                | Case fatality rate HAI AMR                          | 13.8% (8% to 30%)                                                                                                    | Serra-Burriel et al (2020)                                      |
| Value of a premature death                                     | Value of a statistical life                         | US\$4,113 to US\$38,719 (US\$18,706 to US\$112,449) <sup>13</sup>                                                    | Calculation based on Banzhaf (2022)                             |

Mortality cost is estimated based on the number of HAIs annually multiplied by the case fatality rate of HAIs multiplied by the value of a human life. The value of a human life is based on the value of statistical life from the latest meta-analysis from the USA (US\$8 million) converted to SSA using the difference in GDP per capita and an income elasticity of 1.5.

The outputs generated include the annual cost of HAIs, including associated AMR, consisting of the direct healthcare costs, productivity losses, and premature mortality. The annual health care cost is presented as a percent of total health spending, and the annual total cost of HAIs is presented as a percent of GDP. Healthcare and total costs are also estimated per population (in entire country) and per admission.

A one-way sensitivity analysis was conducted by varying input values for each variable shown in Table 2. Where indicated, a lower value was not used because the baseline value is already considered conservative. Justification and references for these values are provided in Annex 1.

**Table 2. Baseline and low/high values for selected variables in sensitivity analysis**

| Variable                                                      | Low (conservative) | Baseline                                                                        | High                                                                            |
|---------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| HAI rate                                                      | Half (6.4%)        | 12.8%                                                                           | Double (25.6%)                                                                  |
| AMR rate                                                      | Half (25%)         | 50%                                                                             | 50% higher (75%)                                                                |
| Cost per hospital bed day                                     | Primary hospital   | Secondary hospital                                                              | Tertiary hospital                                                               |
| Costs of drugs, supplies, laboratory                          | -                  | Mozambique study (district hospital)                                            | Ethiopia study (tertiary hospital)                                              |
| HAI excess length of stay (impacts AMR and productivity cost) | 3 days             | 5 days                                                                          | 10 days                                                                         |
| Value of productive time                                      | -                  | Value added per worker/year in agriculture                                      | Value added per worker/year in industry                                         |
| Case fatality rate                                            | 4.0%               | 6.9%                                                                            | 15.0%                                                                           |
| Value of life                                                 | -                  | Extrapolated from USA based on difference in GDP using income elasticity of 1.5 | Extrapolated from USA based on difference in GDP using income elasticity of 1.2 |

Several costs associated with HAIs as identified by Morel et al (2020) are excluded, either because they are unlikely to be saved in the short- to medium-term (e.g. surveillance and containment programs, education programs and public health or information campaigns) or because they are difficult to quantify and few or no estimates exist in the literature (e.g., cost of unoccupied beds due to isolation, funeral costs and psychological impacts). Treatment of HAIs in healthcare workers are excluded due to lack of estimates. Overall, estimated costs are conservative, drawing on average unit costs that would be typical at secondary hospital level and not considering the additional costs of complications, long-term ICU stay, long-term care and rehabilitation, disability benefits, and expensive pharmaceuticals which might be required in some cases. For AMR, the costs of second-line drugs are used.

## Averted HAI cases and deaths from delivering basic WASH services

High quality intervention studies that isolate the impact of WASH interventions on HAIs are rare, especially in an African context. Studies from SSA and from the rest of the world show variation in effectiveness of WASH interventions, from significant impact (above 50%), to marginal impact (below 40%), to no impact. The impact that improved WASH has on HAI numbers depends on the baseline HAI rates, main transmission pathways and patient risks, and the intensity of and compliance with the intervention. Table 3 presents a summary of selected key literature, in particular systematic reviews, supplemented with intervention studies from SSA. More than half of all cases of health care-associated sepsis are thought to be preventable through safe WASH services and appropriate IPC measures (WHO, 2020; Rudd et al, 2020). Shreiber et al (2018) suggest a sustained potential for the significant reduction of HAI rates in the range of 35%–55% associated with multifaceted interventions, irrespective of a country's income level. Infection control measures during norovirus outbreaks are estimated to be associated with 0.6 (95% CI: 0.3–1.1) times smaller patient case counts (Adams et al, 2022).

**Table 3. Select studies presenting the impact of WASH interventions on HAIs**

| Author (year)             | Type of publication                           | Studies included                              | HAI reduction achieved                                                                                                                                                                                                                                                              |
|---------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Watson et al (2019)       | Systematic review                             | 3 cohort studies                              | 39% to 50%                                                                                                                                                                                                                                                                          |
|                           |                                               | 27 before-and-after studies                   | Statistically significant reduction in at least one HAI in 26 studies (full details in article supplementary materials)                                                                                                                                                             |
| Schreiber et al (2018)    | Systematic review                             | 144 studies                                   | The pooled incidence rate ratios associated with multifaceted interventions were 0.543 for CAUTI, 0.459 for CLABSI, and 0.553 for VAP. The pooled rate ratio was 0.461 for interventions aiming at SSI reduction, and 0.509 for VAP reduction initiatives.                          |
| Umscheid et al (2011)     | Systematic review                             | 64 studies                                    | 65%–70% of cases of CABS and CAUTI and 55% of cases of VAP and SSI                                                                                                                                                                                                                  |
| Luangsanatip et al (2015) | Systematic review                             | 19 studies                                    | The efficacy of the WHO 2005 campaign (WHO-5) reported clinical outcomes in 19 studies. Impacts were consistent with clinically important reductions in rates of infection resulting from improved hand hygiene for some important hospital pathogens.                              |
| Adams et al (2022)        | Systematic review                             | 102 papers describing 162 norovirus outbreaks | Control measures were implemented in 118 (73%) outbreaks and were associated with 0.6 (95% CI: 0.3–1.1) times smaller patient case counts and 0.7 (95% CI: 0.4, 1.0) times shorter durations in hospitals.                                                                          |
| Dramowski et al (2022)    | Intervention study                            | 1 intervention study from South SSA           | Neonatal bloodstream infection rate declined following implementation of the NeoCLEAN intervention, from 6.7 to 3.9/1,000 patient days ( $p = 0.166$ ).                                                                                                                             |
| Aiken et al (2012)        | Intervention studies on SSI prevention in SSA | 24 studies from 9 SSA countries               | Several studies indicated that alcohol-based hand rubs could provide a low-cost alternative to traditional surgical hand-washing methods.                                                                                                                                           |
| Fenny et al (2022)        | Intervention study of alcohol hand rub        | 1 study from Ghana                            | Neonatal bloodstream infection-attributable deaths decreased by 73% and extra length of hospital stay (LOS) decreased by 50%. The post-intervention assessment revealed the ABH program contributed to 16% decline in the incidence of neonatal bloodstream infections at the NICU. |

Based on these findings, the present study uses a baseline reduction in HAIs (and associated case fatality) of 50%, with a range of between 25% to 75% reduction. As presented in Watson et al (2019), before-and-after studies have reported higher percentage reductions in HAIs from the baseline, following prolonged implementation of the WASH measures. Hence, a baseline estimate of 50% reduction in HAIs is likely to be conservative.

### Cost of WASH and HCWM interventions

The study draws on Chaitkin et al, 2022 for costs to achieve basic WASH and waste management in healthcare facilities in 46 low-income countries. These include costs for water supply, sanitation, hygiene and HCWM, as per WHO/UNICEF Joint Monitoring Programme definitions for SDG monitoring. Costs are estimated from the provider perspective and include upfront investments (capital costs), annual operations and periodic maintenance costs for hospitals and non-hospitals in both urban and rural settings.<sup>14</sup> Eight out of the nine focus countries were included in these 46 countries (Kenya excluded). The average cost data for 46 countries<sup>15</sup> from Chaitkin et al (2022) were updated to 2022 values using the GDP deflator for each country.

### Benefit-cost ratio

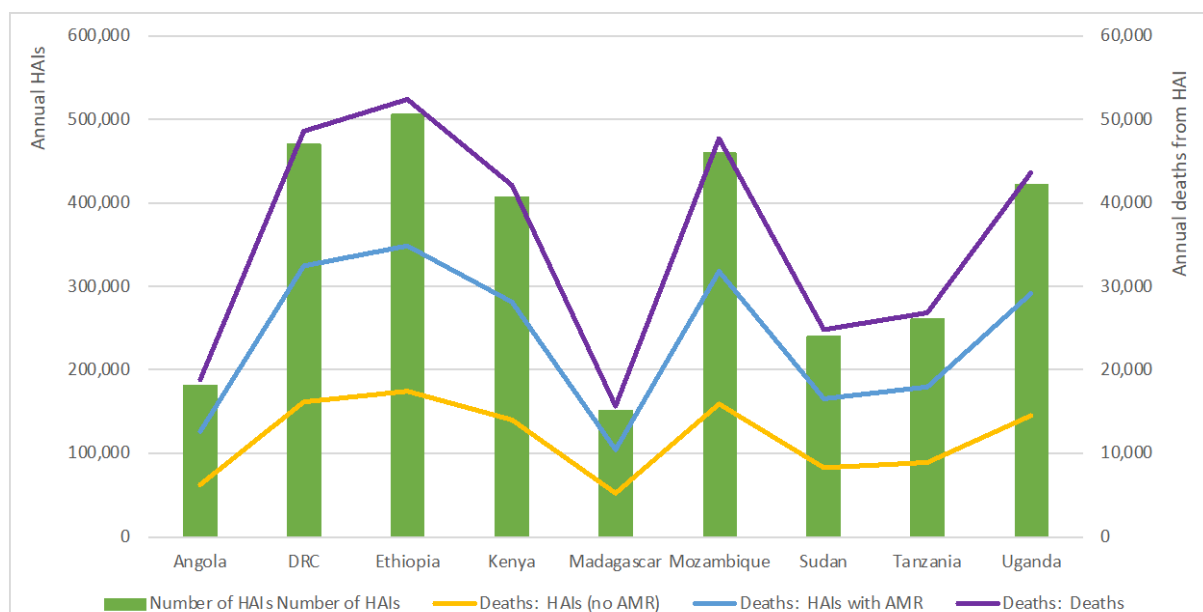
The benefit-cost ratio is calculated as the average economic savings per capita (across the entire population) divided by the average cost per capita of achieving universal coverage of basic WASH in healthcare facilities across nine countries.

### 3. Results

#### Disease burden due to healthcare associated infection.

The total number of HAIs is estimated to be 3.1 million in 2022 in 9 countries of Eastern and Southern Africa, as shown in Figure 4. The number of deaths resulting from these HAIs is estimated to be 320,000 in 2022. The number of HAIs and deaths is closely related to the population size, and hence the number of hospital admissions, with the highest numbers in Ethiopia and Mozambique. That said, drawing on official government data, admissions per 100 population vary widely from 2.5 per 100 population in Ethiopia to 11.2 per 100 population in Mozambique, hence explaining some of the differences in HAI costs.

**Figure 4. Estimated number of healthcare associated infection and related deaths in 9 countries in Eastern and Southern Africa**

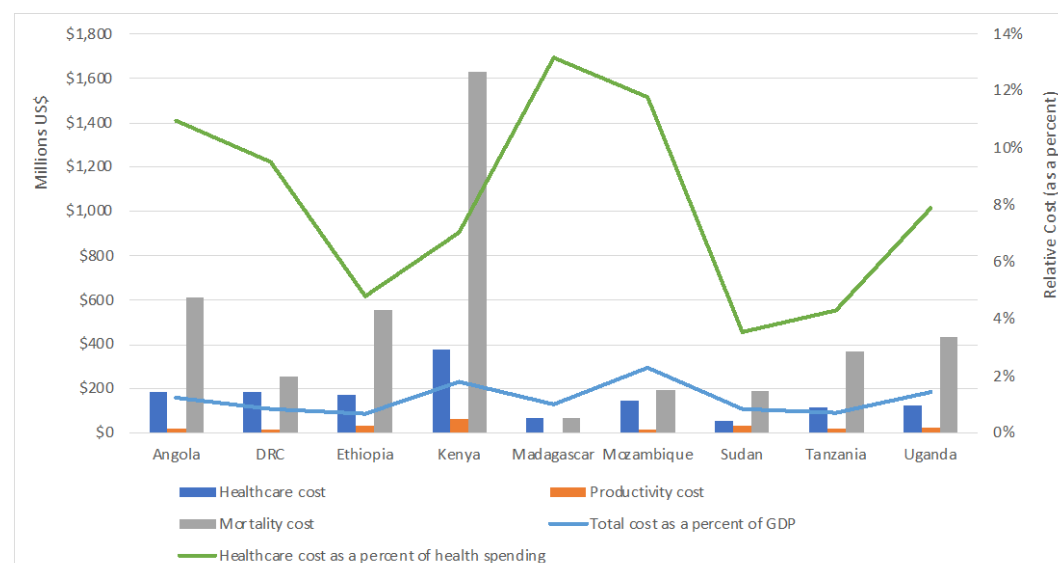


Source: author estimates

#### Economic impacts due to HAIs

Economic estimates were aggregated from the healthcare costs, the productivity losses and the premature deaths associated with HAIs, including AMR. The total economic losses are at least US\$6 billion in 2022 across the 9 countries. As shown in Figure 5, the costs of premature death account for 72% of the total costs, due to the relatively high case fatality rate from HAIs. In Kenya, total costs were US\$2.1 billion in 2022, and US\$820 million in Angola. Costs as a percent of GDP are highest in Kenya and Mozambique, at 1.8% and 2.3%, respectively. Across the nine countries, total cost as a percent of GDP is 1.14% in 2022, and healthcare cost as a percent of total health expenditure is 7.1%.

**Figure 5. Estimated healthcare costs, productivity costs and mortality costs resulting from HAIs in nine countries, and costs as a percent of GDP**



Source: author estimates

## Sensitivity analysis

The sensitivity analysis conducted on 6 variables with inherent uncertainty indicates that the cost of illness varies to a moderate degree (Figure 6). The analysis is limited by the fact that input values were varied one at a time, and not in combination. However, combining high (or low) values of several input variables together would generate extreme values which are unlikely to occur in practice. The baseline results are most affected by assumptions for (a) HAI rates, which underly the estimate of all the impacts, on (b) case fatality rates; and on (c) the value of life. The latter two have the greatest impact because premature death accounted for 72% of the impacts under the baseline. The value of time had the greatest impact on the productivity costs. Overall, across the nine countries, the baseline cost of illness of US\$6 billion was most affected by (a) value of life assumption, which led to more than a doubling of the total costs to over US\$16.4 billion, and (b) lower CFR, HAI and AMR rates which led to reduction in total costs to less than US\$5 billion.

## Cost-benefit analysis

Across the nine countries, the annual healthcare costs per capita (across the entire population) was US\$2.75, or US\$2.04 per capita for the low-income countries.<sup>16</sup> This compares with a total economic cost of US\$11.5 per capita for all 9 countries, or US\$7 per capita for the low-income countries. The costs of providing basic WASH, HCWM and environmental cleaning in low-income countries is estimated at US\$0.81 per capita per year (US\$0.30 for capital cost and US\$0.51 for recurrent cost when universal coverage is achieved).<sup>17</sup> Assuming 50% reduction in HAIs through the provision of basic WASH, HCWM and environmental cleaning the resulting benefit-cost ratios are 1.22 for financial health care savings alone (1.38 with middle income countries included), and 4.2 for all economic costs (5.8 with middle income countries included)(see Table 4).

**Table 4. Estimation of benefit-cost ratios for WASH and HCWM interventions**

| Countries included | Cost-of-illness per capita |       | Averted costs (50% reduction) |       | Intervention costs | Benefit-cost ratio |       |
|--------------------|----------------------------|-------|-------------------------------|-------|--------------------|--------------------|-------|
|                    | Health care                | Total | Health care                   | Total | Health care        | Health care        | Total |
| Low-income         | 2.04                       | 7.0   | 1.02                          | 3.5   | US\$0.83           | 1.22               | 4.2   |
| All 9 countries    | 2.75                       | 11.5  | 1.38                          | 5.8   | US\$1.00           | 1.38               | 5.8   |

Source: author estimates. Intervention costs from Chaitkin et al (2022).

## 4. Discussion

This study has demonstrated that there are major hidden costs of HAIs in SSA. Every day, millions of Africans receive inpatient care in a healthcare facility – whether for a chronic illness or acute disease, surgical procedures or for labor and delivery and postpartum services – but in their vulnerable state an estimated 10% or more of these people become infected by a microorganism while receiving medical care, a high proportion of which are resistant to first, second- and even third-line drugs.

By quantifying the major costs associated with these infections at national level – direct healthcare costs of treatment, productivity losses for the patients, and the value of lost lives – this study shows that the economic costs exceed 1.14% of the GDP of nine countries in Eastern and Southern Africa and the direct healthcare costs exceed 7% of total health spending in these countries.

While efforts were made to select data sources that best reflect the national contexts in the countries of study, there are still some methodological weaknesses, excluded impacts and data limitations associated with a multi-country desk study that is reliant on secondary data sources.

This study adopted a standard cost-of-illness methodology (Morel et al, 2020; Shrestha et al, 2018). While previous studies have assessed the direct healthcare costs of HAIs in SSA (Aerts et al, 2022; Schäferman et al, 2020; Gidey et al; 2023; Otieku et al, 2023), none have included the indirect costs to the patient. Hence, standard economic approaches were used to value the loss of productive time due to a longer time spent away from work or home. In some instances, this loss may be incurred by a caregiver or family member– for example, a woman who has just delivered a baby is unlikely to return to work immediately, but the HAI she contracts may result in a loss of earnings or productive time for a partner or relative.

Quantifying the cost of premature death due to HAI is important, given that a significant proportion of HAIs lead to premature death, especially for AMRs when a more expensive second- or third-line drug is unavailable or unaffordable. This study used value-of-statistical-life (VSL) methodology to quantify the value of life. It did not value disability-adjusted life-years (DALYs) lost due to the lack of data on the age of the patient (and age at death). However, due to the lack of VSL studies in the study countries, VSL was estimated using a meta-analysis of studies from the USA and adjusted for the differences in economic development (GDP per capita) and using a conservative income-elasticity of 1.5. The latter allows for the likelihood that people have a lower willingness to pay for interventions that prevent death when their income is lower. Sensitivity analysis showed that when case fatality increased from 6.9% to 15% and when income elasticity of 1.2 was used instead of 1.5, that the total economic losses increased from US\$6 billion to US\$11.0 billion and US\$16.4 billion, respectively.

Several quantifiable costs of HAIs - and specifically of AMR HAIs - were excluded due to lack of available data and difficulties in aggregating to the national level. These include:

- The costs of surveillance and containment programs, education programs and public health or information campaigns related to HAIs.
- Additional costs of longer-term complications of HAIs, as it was assumed that the initial cost of treatment would lead to either discharge or death. However, some patients may require long-term ICU stay, long-term care and rehabilitation, and disability benefits.



BRIEF COMMUNICATION

Open Access



# Socioeconomic factors influencing rural-urban ambulance response time disparities in Connecticut

Eashwar Krishna<sup>1\*</sup>

## Abstract

Across the U.S, it is a documented fact that rural areas have longer ambulance response times and tend to have lower median income. The objective of this study was to test if the rural-urban emergency medical service (EMS) response time disparity was related to wealth disparity in the state of Connecticut. All mean EMS response times were sourced from the 2016 Office of Emergency Medical Services Data Report. Rural definitions were sourced from the Connecticut Office of Rural Health. Median income data was drawn from the Connecticut Office of Policy and Management. A Mann-Whitney U test determined if the average rural EMS response time was greater than the non-rural EMS response time. Pearson coefficients quantified the relationship between median income and EMS response time. A t-test ascertained if the average median income differed between the two datasets. The mean EMS response time was 12.98 min (SD = 3.36) rural and 8.26 min (SD = 2.12) non-rural. Rural mean response time and median income were not significantly correlated ( $r = -.148, p = .247$ ); non-rural mean response time and median income were also not significantly related. No significant disparity was detected ( $t = 0.478, p = .633$ ) between the mean rural household income (\$98,258) and mean non-rural household income (\$95,706). Significant disparities in EMS response times can exist between rural and non-rural towns separate from median income trends, as is the case in Connecticut. These findings may have limited generalizability because of Connecticut's relatively high median income as compared to other states yet may be relevant to states with similar economic metrics.

**Keywords** Rural health, Emergency medical services, Ambulances, Connecticut, Public Health

## Introduction

It has been demonstrated that shorter ambulance response times improve immediate and long-term outcomes during cardiac arrests [1, 2]; for example, O'Keefe et al. [3] found for every one minute reduced in response time, the chance of survival increased by 24%. While patient survival rates of an adverse cardiac event can be greatly improved through bystander education and

intervention, the same cannot be said for other out-of-hospital medical emergencies. One such case is foreign body airway obstruction events: in only about 5–13% of cases are bystanders able to make a correct diagnosis and perform the appropriate maneuver; moreover, approximately 30–40% of all such events occur when the victim is alone [4]. On average, the victim has six minutes before they begin to suffer from hypoxia-related brain damage.

As such, it is a public health imperative to provide timely professional care to the acutely ill and injured. Mell et al. [5] report that across the USA, rural response times are almost double suburban and urban responses. A good amount of heterogeneity among the results of

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rural-urban response time disparity studies has also been documented [6], creating the need to study various regions of the United States in isolation from each other.

While the relationship between median income and response times [7], as well as rurality and response times [5], has been studied, the interrelation of these two factors has not yet been the subject of sufficient research. Connecticut, a northeastern state of the USA, is an excellent case study to understand the unique challenges that face rural EMS agencies and the role, if any, of median income. On a national level, urban counties tend to have higher median income than rural counties [8]; in contrast, Connecticut has the highest median income and lowest poverty rates for rural areas in the country [9], and possibly no significant rural-urban wealth disparity.

I hypothesized that a significant disparity between rural and non-rural response times will persist in Connecticut; however, I expected that this disparity would not be correlated with median income.

## Methods

All mean EMS response times were sourced from the 2016 Office of Emergency Medical Services Data Report, published by the Connecticut Department of Public Health. Towns, neighborhoods, or villages with fewer than thirty recorded EMS encounters, incomplete ZIP (Zone Improvement) code data, or incomplete time point data were excluded from the report [10]. It should be noted that many of these excluded areas were in fact neighborhoods under the jurisdiction of a recorded town. The only independent rural town without a mean response time reported due to insufficient reports was Lyme; Union, North Canaan, Sharon, and Warren were excluded from the report with no explicit explanation yet it may be assumed that they lacked complete time points or ZIP code data. Stafford, Griswold, and Killingly – classified for the purposes of this study as non-rural – were also excluded from the report due to incomplete data.

The Connecticut Office of Rural Health has published a list of towns considered rural according to the state definition, which was adopted for use in this study; other rural definitions such as RUCC and RUCA were not applicable to this study as national standards do not use towns as their fundamental units of study. Towns were chosen as the focus of this study because they are the smallest units of governance in Connecticut, and as such EMS policy and median income can vary widely between even neighboring towns in the same county.

All towns with a population of fewer than ten thousand residents as well as a population density of five hundred people or fewer per square mile were classified as rural [11], yielding 68 rural towns in total; 63 towns were used for data analysis due to the remaining five lacking data. Urban and suburban towns were grouped together since

the nationwide mean times for both groups were highly similar; in total, 95 towns and cities were used for the non-rural comparison group. The dataset used in the analysis can be found in Appendix A and B.

Average household median income data was drawn from the Municipal Fiscal Indicators dataset released by the Connecticut Office of Policy and Management [12]. Median income and mean response time data were paired for each town, and the overall dataset was split into rural and non-rural sets. Only towns being used for the response time comparison were included in these datasets; the eight towns excluded for the aforementioned reasons were also excluded from the median income calculations. The median income sets can be found in Appendix C and D.

## Statistical analysis

The null hypothesis was that the distribution of mean response times (MRT) for rural and urban towns would be equal, whereas the alternative hypothesis was that the distributions would not be equal. A two-sample Wilcoxon-Mann-Whitney U test with  $(\alpha)=0.05$  was performed to test if the rural distribution of response times was stochastically different than the urban distribution. The two datasets had different distribution shapes, and thus a non-parametric test was necessary. Measures of central tendency were calculated and the Tukey fence method was used to better describe the data and discover outliers, respectively.

Pearson coefficients of correlation were calculated for the relationship between average household median income and mean response time for both sets. Furthermore, a two-tailed t-test with  $(\alpha)=0.05$  was performed to determine if median household income varied between the rural and non-rural datasets. The data were analyzed using IBM SPSS statistics (Version 29).

## Ethics approval

This study only analyzed publicly-available, de-identified data and as such did not meet the criteria for IRB review according to 45 CFR 46.102.

## Results

A total of 158 towns were included in the analysis. The mean response time for the rural dataset was 12.98 min ( $SD=3.6$ ), while the same for the non-rural data was 8.26 min ( $SD=2.12$ ); the distributions in the two groups differed significantly ( $z=8.3464$ ,  $p<.001$ ). Median response times were 12.8 min and 7.7 min for rural and non-rural respectively. The observed standardized effect size was medium at 0.66, meaning that the magnitude of the difference between the two datasets was sizable. No outliers were observed in the rural data; Simsbury and Weston were non-rural dataset outliers with 15.8- and

14.6-minute mean response times, respectively, yet were still included in the analysis.

Neither rural mean response time and median income ( $r = -.148$ ,  $p = .247$ ) nor non-rural mean response time and median income ( $r = .137$ ,  $p = .185$ ) were found to be significantly correlated. There was no significant difference in average household median incomes between the two datasets ( $t = 0.478$ ,  $p = .633$ ).

## Discussion

While rural response times statewide are lower than the national rural average, a significant disparity still exists between non-rural and rural towns within Connecticut; this inequity has serious implications for patient outcomes. Though only a few urban/suburban towns have a mean response time below five minutes, generally considered the gold standard, the average non-rural MRT is not significantly far from it. In contrast, the average rural response time in Connecticut is almost thirteen minutes, well over double the time necessary to consistently save victims of traumatic injuries, choking, or other severe events.

In addition, the data makes it evident that median income has no statistically significant impact on response time within either dataset. Moreover, there is no significant wealth disparity as measured by household median income between rural and non-rural areas. This suggests that more focus should be directed towards other aspects of rural EMS, such as staffing shortages and town policies, rather than solely focusing on socioeconomic factors. It should be noted that this conclusion is specific to Connecticut; it may or may not be the case elsewhere that the most pertinent issue to response time disparity is a rural-urban wealth disparity. While lower-income towns may struggle with EMS procurement and operational costs, which can be unpredictable [13], more than higher-income towns, multiple papers have demonstrated that the impact of non-financial factors may be more influential in resolving the response time disparity. For example, Wei Lam et al. found that changes to dispatch policy and more effective allocation of existing ambulances would be more effective than merely adding more ambulances in reducing response times overall [14]. As for rural response times specifically, poor access to primary care for non-acute health management, provider burnout, and lack of opportunities for provider-initiated workplace improvements have been cited by Newton et al. as issues contributing to lengthy rural EMS response times [15]. The same study indicated that directing federal and state funding to community paramedicine initiatives may be effective at managing the underlying health disparities present in rural and remote communities; failure to provide consistent primary care to these populations tends

to result in increased EMS utilization [16], which can put already-strained units out-of-service.

Other communities may find it beneficial to conduct small-scale, local investigations regarding EMS response in their jurisdiction. Much of the existing work on response time and rurality focuses on county [17] or ZIP codes [5] as geographic units of analysis; however, this is frequently not the level at which policy and quality improvement interventions are implemented. For example, in the New England region of the United States, which includes Connecticut, municipalities such as villages, towns, and cities regulate emergency services independently, with limited oversight from the state. The implication of this fact is that county-focused analysis may be erroneously grouping together a vastly heterogeneous number of local EMS services, with varying degrees of funding, privatization, and quality improvement initiatives. Conversely, ZIP code-focused analysis may become too granular, as a single municipality frequently covers multiple ZIP codes. This study attempts to illustrate municipality-level differences, which emphasize the necessity of local policy innovations to combat long response times.

Such policy changes are already underway in Connecticut, and this study underscores their necessity and potential impact; since rural-urban response time disparities in the state do not appear to be correlated with median income, such disparities are more remediable by tailoring solutions to the unique challenges faced by rural communities. Rather than utilize limited funds to solely subsidize equipment purchases, it may be far more impactful to pursue EMS innovation and fundamental alterations to the existing pre-hospital care infrastructure. A prime example of such a project is the Mobile Integrated Health workgroup, chaired by the Connecticut Department of Public Health, which aims to create a community paramedicine initiative that will integrate with existing EMS systems. Notably, the group's most recent report indicates a goal to "triage and connect non-urgent 9-1-1 callers with relevant caregivers and assistance instead of dispatching an ambulance crew" [18]. The results of this study in combination with existing research indicate that it will be efficacious for additional attention to be given to community-level interventions in rural areas such as health literacy programs, increased incentives for primary care providers, and increased expansion of mobile integrated health solutions in Connecticut.

## Limitations

In order to accurately capture the effect of median income and town policy, analysis was conducted by town lines; however, the source data for mean response time did not classify towns by strict jurisdictional borders, and as such included neighborhoods or villages of towns

as separate entries. In these situations, the town center data point was used for mean response time; as such, the fidelity of the data is a limitation of this research. This study also limited its analysis of socioeconomic factors to median household income, and other such factors may affect response times.

The data utilized in this study, collected in 2016, may present certain implications for the results. The age of the data is a critical factor to consider, as it may not fully capture recent changes in EMS infrastructure, technological advancements, or policy shifts that have occurred since then. For instance, while rural towns have predominantly relied on volunteer services for coverage, a declining number of volunteers has led some of these towns to contract private agencies. Subsequently, these municipalities reported decreased response times [19]. Such trends, if significant across most towns included in the rural dataset, may be hypothesized to decrease the disparity in response time observed in this study.

Despite these shifts, a more recent report from the Department of Public Health (DPH) indicates that the overall environment of EMS as it relates to rural-urban response time disparities has remained roughly the same. At the time of the analysis, the non-rural mean response time (MRT) was 8.26 min; this report was published five years later and reports a suburban MRT of 8.12 min and an urban MRT of 7.39 min [20]. Rural towns were noted to have a mean response time of 10.07 min in the more recent report, compared to 12.8 min in this study. This disparity can be explained by data reporting standards changing; namely, at the time of the data analysis in this study, median response times were not stratified by rural status by the DPH. In the more recent report, several towns that were previously under the non-rural dataset have been included in the rural MRT average, which understandably lowers this number. While this phenomenon is partially explained by population trends and possibly differences in methodology, a repetition of the analysis in this study with newer data would be warranted if town-specific conclusions need to be drawn. However, the general trend illustrated by this study and the interpretation thereof remains relevant to both the state of Connecticut as well as other communities who might benefit from investigating novel solutions to the indisputable disparity in rural pre-hospital care.

## Conclusion

There exists a significant rural-urban disparity in ambulance response times in Connecticut which is decoupled from median income, a concerning trend that demands amelioration in the near future to ensure equitable access to timely healthcare. The most effective strategy to do so will be one that focused on policy and innovation in both

emergency medical services as well as care focusing on chronic and preventable conditions.

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s43999-024-00055-9>.

Supplementary Material 1

## Funding

No funding was received to conduct this study or assist with the preparation of this manuscript.

## Data availability

Data was publicly available through the Connecticut Department of Public Health.

## Declarations

### Ethical approval and consent to participate

This study did not qualify to be reviewed by an IRB.

### Conflict of interest

The author reports no conflict of interest for this study.

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## References

1. Bürger A, Wnent J, Bohn A, Jantzen T, Brenner S, Lefering R et al (2018) The effect of ambulance response time on survival following out-of-hospital cardiac arrest: an analysis from the German resuscitation registry. *Deutsches Ärzteblatt International* 115(33–34):541–548. <https://doi.org/10.3238/arztebl.2018.0541>
2. Holmén J, Herlitz J, Ricksten S, Strömsöe A, Hagberg E, Axelsson C et al (2020) Shortening ambulance response time increases survival in out-of-hospital cardiac arrest. *J Am Heart Association* 9(21):e017048. <https://doi.org/10.1161/JAHA.120.017048>
3. O'Keeffe C, Nicholl J, Turner J, Goodacre S (2011) Role of ambulance response times in the survival of patients with out-of-hospital cardiac arrest. *Emerg Med J* 28(8):703–706. <https://doi.org/10.1136/emj.2009.086363>
4. Luczak A (2019) Effect of body position on relieve of foreign bodies from the airway. *AIMS Public Health* 6(2):154–159 Published 2019 Apr 24. <https://doi.org/10.3934/publichealth.2019.2.154>
5. Mell HK, Mumma SN, Hiestand B, Carr BG, Holland T, Stopyra J (2017) Emergency Medical Services Response Times in Rural, Suburban, and Urban Areas. *JAMA Surg* 152(10):983–984. <https://doi.org/10.1001/jamasurg.2017.2230>
6. Alruwaili A, Alanazy ARM (2022) Prehospital Time Interval for Urban and Rural Emergency Medical Services: a systematic literature review. *Healthc (Basel)* 10(12):2391 Published 2022 Nov 29. <https://doi.org/10.3390/healthcare1012391>
7. Hsia RY, Huang D, Mann NC, Colwell C, Mercer M, Dai M et al (2018) A US National Study of the Association between Income and Ambulance Response Time in Cardiac arrest. *JAMA Netw Open* 1(7):e185202 Published 2018 Nov 2. <https://doi.org/10.1001/jamanetworkopen.2018.5202>
8. Bureau UC, Differences in Income Growth Across U.S. Counties (2018). The United States Census Bureau. Published December 6, <https://www.census.gov/library/stories/2018/12/differences-in-income-growth-across-united-state-s-counties.html>
9. New Census Data Show Differences Between Urban and Rural Populations (2016) The United States Census Bureau. Published Dec 30, <https://www.census.gov/newsroom/press-releases/2016/cb16-210.html>
10. Kloter A, Coler R (2017) Office of Emergency Medical Services Data Report. Connecticut Department Public Health; Published Dec 13, <https://portal.ct.gov/-/media/Departments-and-Agencies/DPH/dph/ems/pdf/CEMSTARS/2016 OEMSAnnualDataReportpublic.pdf>